

LLM-as-a-Judge Report: Self-doubt GSM8K_20000

What was measured?

The judge scores visible self-questioning in the reasoning trace, independently of whether the final answer is correct. A score of 0 means a straight-line trace with no hesitation; 10 means pervasive uncertainty, reversals, or looping. The judge returns the score, a verbatim evidence quote when available, and a one-line summary. The tables below use 0–10 scores and include only valid answered traces from `gsm8k_20000_judge_traces.jsonl`.

Examples at different scores

The examples below show the score, final-answer outcome, the judge’s evidence, and its one-line summary. An empty evidence field means that the trace contained no suitable verbatim cue for that dimension.

Trace `typo50_real40`, index 50. **Score:** 0 on the 0–10 scale. **Final-answer outcome:** wrong.
Judge evidence:

I don’t see any mistakes in my reasoning here.

One-line summary: The trace led to the scores due to a clean straight-line solution with no noticeable self-doubt or misunderstanding of the corrupted words.

Trace `typo25_real40`, index 2. **Score:** 2 on the 0–10 scale. **Final-answer outcome:** correct.
Judge evidence:

I was initially confused whether the 150% increase was based on the total cost or the original price

One-line summary: The trace shows a brief hesitation in calculating the increase in value, but overall understands the problem and corrupted words correctly.

Trace `typo50_real40`, index 0. **Score:** 5 on the 0–10 scale. **Final-answer outcome:** correct.
Judge evidence:

Wait, does that mean she uses four eggs each day to bake muffins? Or is it four eggs per something?

One-line summary: The model’s hesitation and questioning of the wording, such as ‘Wait, does that mean she uses four eggs each day to bake muffins?’ led to the scores.

Trace typo25_real70, index 0. **Score:** 8 on the 0–10 scale. **Final-answer outcome:** wrong.
Judge evidence:

Wait, maybe the problem is that she lays 16 eggs per azy, which is 16 eggs per day.

One-line summary: The trace’s repeated reconsideration of the meaning of ‘azy’ and its potential typos led to high self-doubt and misunderstanding of the question’s intended meaning.

Aggregate tables

Table 1: Mean score by real-word typo ratio

real_ratio	n	self_doubt_score_mean
10	1449	5.206
40	1474	5.288
70	1454	5.12

Table 2: Mean score by typo percentage

typo_percentage	n	self_doubt_score_mean
0	485	4.495
25	1466	4.862
50	1472	5.12
75	1439	5.641

Table 3: Mean score by final-answer correctness

outcome	n	self_doubt_score_mean
correct	3816	4.691
wrong	1046	6.753

Consequences

In this run, self-doubt decreases as the real-word ratio increases: the mean is 1.001 at 10 percent real-word typos, 0.889 at 40 percent, and 0.826 at 70 percent. This is consistent with the interpretation that non-real-word typos are more visibly strange and can trigger more doubt, while real-word typos can be read fluently and silently.